

实测天体物理作业

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2022 年 2 月 28 日

1. 单位球面的总面积为：

$$S = \iint_{\Omega} \sin \theta \, d\theta d\varphi$$

以球面度为单位时：

$$\begin{aligned} S &= \iint_{\Omega} \sin \theta \, d\theta d\varphi \\ &= \int_0^{\pi} \sin \theta \, d\theta \int_0^{2\pi} d\varphi \\ &= (-\cos \theta)|_0^{\pi} \cdot \phi|_0^{2\pi} \\ &= 4\pi \text{ sr} \end{aligned}$$

以平方度为单位时：

$$\begin{aligned} S &= \iint_{\Omega} \sin \theta \, d\theta d\varphi \\ &= \int_0^{180} \sin \theta \, d\theta \int_0^{360} d\varphi \\ &= \left(-\frac{180}{\pi} \cos \theta\right)\bigg|_0^{180} \cdot \phi|_0^{360} \\ &= \frac{129600}{\pi} \text{ deg}^2 \\ &\approx 41253 \text{ deg}^2 \end{aligned}$$

可见有：

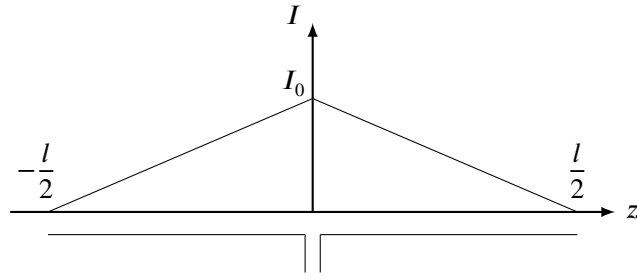
$$1 \text{ sr} = \frac{32400}{\pi^2} \text{ deg}^2 = \left(\frac{180}{\pi}\right)^2 \text{ deg}^2 \approx 3282.8 \text{ deg}^2$$

由于观测分辨率为 $1 \text{ beam} = 1 \text{ arcmin}^2$ ，且 $1 \text{ deg}^2 = 60^2 \text{ arcmin}^2 = 3600 \text{ arcmin}^2$ ，故：

$$1 \text{ sr} = 3600 \cdot \frac{129600}{\pi} \text{ beam} = 1.1818 \times 10^7 \text{ beam}$$

2. 设偶极天线的长度为 l ($l \ll \lambda$)，中心处最大电流为 I_0 ，则沿天线上的电流分布近似为线性形式：

$$I(z) = I_0 \left(1 - \frac{2}{l}|z|\right), \quad |z| \leq \frac{l}{2}$$



设天线的电偶极矩为 $\mathbf{p} = p\mathbf{e}^{-i\omega t}\hat{\mathbf{z}}$, 则有:

$$\frac{\partial \mathbf{p}}{\partial t} = \iiint_V \mathbf{J}(\mathbf{r}) dV = \int_{-l/2}^{l/2} I(z) dz = \frac{1}{2} I_0 l, \quad \mathbf{p} = \frac{i}{2\omega} I_0 l \mathbf{e}^{-i\omega t} \hat{\mathbf{z}}$$

其产生的辐射场为:

$$\mathbf{E} = \frac{p\omega^2}{4\pi\epsilon_0 c^2} \sin\theta \frac{e^{ikr}}{r} \hat{\boldsymbol{\theta}} = \frac{i\omega I_0 l}{8\pi\epsilon_0 c^2} \sin\theta \frac{e^{-i(\omega t - kr)}}{r} \hat{\boldsymbol{\theta}}$$

$$\mathbf{H} = \frac{p\omega^2}{4\pi c} \sin\theta \frac{e^{ikr}}{r} \hat{\boldsymbol{\phi}} = \frac{i\omega I_0 l}{8\pi c} \sin\theta \frac{e^{-i(\omega t - kr)}}{r} \hat{\boldsymbol{\phi}}$$

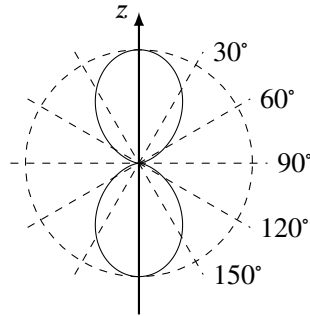
则 Poynting 矢量为:

$$\mathbf{S} = \frac{1}{2} \mathbf{E} \times \mathbf{H}^* = \frac{\mu_0 c}{32} \left(\frac{I_0 l}{r\lambda} \right)^2 \sin^2\theta \hat{\mathbf{r}}$$

其功率的角分布可以从 Poynting 矢量得到:

$$P_n = \frac{dP}{d\Omega_n} = \lim_{r \rightarrow \infty} \hat{\mathbf{r}} \cdot (r^2 \mathbf{S}) = \frac{\mu_0 c}{32} \left(\frac{I_0 l}{\lambda} \right)^2 \sin^2\theta$$

则有: $P_{max} = \frac{\mu_0 c}{32} \left(\frac{I_0 l}{\lambda} \right)^2$, 归一化的功率方向图 $P_n(\theta, \phi) = \sin^2\theta$, 绘图如下:



天线的总功率为:

$$P = \frac{\mu_0 \pi c}{12} \left(\frac{I_0 l}{\lambda} \right)^2$$

方向增益为:

$$G_n(\theta, \phi) = \frac{4\pi P_n}{P} = \frac{3}{2} \sin^2\theta$$

方向系数 $D = G_{max} = \frac{3}{2}$, 以 dB 为单位表示为:

$$D_{dB} = 10 \lg \frac{3}{2} = 1.761 \text{ dB}$$

天线立体角为:

$$\Omega_A = \frac{4\pi}{G_{max}} = \iint_S P_n d\Omega = \frac{8\pi}{3}$$

3. 流量密度为:

$$S = \frac{P}{4\pi r^2 \Delta\nu} = \frac{500 \text{ W}}{4\pi \times (1 \text{ m})^2 \times 1 \text{ MHz}} = 3.979 \times 10^{-7} \text{ Wm}^{-2}\text{Hz}^{-1} = 3.979 \times 10^{19} \text{ Jy}$$

4. 天线立体角为:

$$\begin{aligned}\Omega_A &= \iint_S P_n d\Omega \\ &= 0.1 \times 2\pi(1 - \cos 10^\circ) - 0.1 \times 2\pi(1 - \cos 1^\circ) + 1 \times 2\pi(1 - \cos 1^\circ) \\ &= 1.04 \times 10^{-2}\end{aligned}$$

主瓣立体角为:

$$\begin{aligned}\Omega_{MB} &= \iint_{MB} P_n d\Omega \\ &= 1 \times 2\pi(1 - \cos 1^\circ) \\ &= 9.57 \times 10^{-4}\end{aligned}$$

主瓣效率为:

$$\eta_B = \frac{\Omega_{MB}}{\Omega_A} \times 100\% = 9.2\%$$